Partial Universalisation in Threshold Public Goods: Actions to Inclinations

Haochen Ni

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Abstract

This paper studies how partial universalisation shapes contribution decisions in a finite threshold public-good environment. An individual evaluates a contemplated strategy under a counterfactual in which each other group member independently adopts that strategy with probability σ, otherwise contributing with an exogenous background probability p. For pure commitments, the contribution premium equals the probability that the contribution-oriented counterfactual reaches the provision threshold while the abstention-oriented one does not, an event we call *collective pivotality*. For every $0 < \sigma < 1$, this premium is strictly single-peaked in p, so contribution is preferred only at intermediate levels of background cooperation under suitable cost-benefit ratios. Extending the choice set to contribution inclinations, we establish that, for every admissible group size and threshold, a non-empty open set of parameters supports a unique interior optimum, even potentially when pure abstention is preferred to pure contribution. Moreover, while stronger counterfactual linkage always raises the pure-contribution premium, it need not increase the optimal inclination. These results distinguish the ranking of deterministic commitments from the choice of contribution inclinations under partial universalisation.

Keywords: Kantian reasoning; homo moralis; collective pivotality; threshold public goods; assortative matching.

1. INTRODUCTION

Much of economic analysis evaluates an action while holding others' behaviour fixed. This typically yields a unilateral-deviation logic, in which contributing to a threshold public good is attractive only to the extent that the individual's own action changes the provision outcome. This is known as the *homo oeconomicus* preference. Yet in many large-scale collective-action settings, including voting, boycotts, and climate action, the probability of unilateral pivotality can be very small. This raises a natural question: how can an individual regard her contribution as decision-relevant when her action is unlikely to change the collective outcome on its own? One possible answer is Kantian reasoning. Rather than holding others' actions fixed, an individual asks what would happen if others acted as she does. At the boundary case of complete universalisation, the individual is called *homo kantiensis*. Related approaches are studied by Laffont (1975) and Roemer (2010, 2015). Complete universalisation, however, is a demanding counterfactual. In many environments, an individual may regard her contemplated action as related to the actions of others without expecting the entire population to act identically. This paper studies such partial Kantian reasoning. A partial Kantian individual evaluates a contemplated action under a counterfactual in which each member adopts that action independently with probability σ, while otherwise acting according to a background contribution probability p. When $\sigma = 0$, the focal individual's action carries no counterfactual linkage to the member's actions, and the resulting premium reduces to ordinary unilateral pivotality. When $\sigma = 1$, she fully universalises her action, recovering the standard Kantian benchmark.

Our analysis builds on the framework of *homo moralis* developed by Alger and Weibull (2013, 2016). They show that assortative matching can favour preferences under which individuals evaluate actions partly through counterfactual profiles in which some of the group members adopt the same action. We take this counterfactual evaluation structure as given and study the decision-theoretic implications it generates in finite threshold public-good environments. Under the conditional-independence specification, each member's action is replaced by the focal individual's contemplated action with probability σ, implying that the number of *like-minded* members follows a binomial distribution. We refer to these members as *like-minded* because

their actions are linked to the focal individual's counterfactual evaluation. Intuitively, they are individuals whom the focal individual treats as sharing the same reasoning process or logic in the relevant counterfactual. More precisely, consider a group of n individuals in which the public good is provided if at least k contribute. Let p denote the background probability that an unlike-minded partner contributes. Under the contribution-oriented counterfactual, each partner contributes with probability $\sigma+(1-\sigma)p$. Under the abstention-oriented counterfactual, each partner contributes with probability $(1-\sigma)p$. For pure commitments, we define the collective-pivotality premium, $\Delta^{PC}_{n,k}(p)$, as the difference between the probability of provision under these two counterfactual profiles. The focal individual prefers pure contribution to pure abstention whenever this premium exceeds the cost-benefit ratio. Under common coupling of the two counterfactuals, the premium admits a direct probabilistic representation. It is the probability that the contribution-oriented counterfactual reaches the provision threshold while the abstention-oriented counterfactual does not. We call this event *collective pivotality*. When no group member is like-minded with the focal individual, collective pivotality reduces to unilateral pivotality. When one or more members are like-minded, the threshold can be crossed by the focal individual together with those like-minded members. The individual therefore evaluates whether the probabilistically linked response generated by her contemplated action changes the collective outcome.

Our first result characterises the geometry of pure commitment. For every partial Kantian linkage $0 < \sigma < 1$, the collective-pivotality premium is strictly single-peaked in the background contribution probability p. When p is low, greater background cooperation raises the value of contribution by moving the group closer to the threshold. When p is high, the same increase lowers the value of the contribution because provision becomes increasingly likely even under the abstention-oriented counterfactual. Consequently, for intermediate cost-benefit ratios, contribution is preferred only over a bounded interval of background contribution rates. Below this interval, even the contribution-oriented counterfactual is unlikely to generate sufficient support; we call this the *disillusionment effect*. Above it, provision is sufficiently likely under abstention that the focal individual's contribution becomes redundant; we call this the *redundancy* or *free-rider effect*. Thus, abstaining from contribution can reflect the opposite behaviour logic.

We then extend the framework beyond deterministic commitments by allowing individuals to choose inclinations, represented by mixed strategies. This environment generates a richer choice structure. Even when pure abstention yields a higher payoff than pure contribution, an interior inclination may be optimal because the individual values the possibility of contributing while maintaining a probabilistic linkage with the behaviour of like-minded members. Thus, partial universalisation does not merely interpolate between unilateral and complete Kantian reasoning; it generates a distinct class of probabilistic decisions.

The paper makes three contributions. First, it develops a general framework for partial Kantian, or partial universalisation, reasoning in threshold environments that encompasses both pure commitments and mixed inclinations. Second, it identifies collective pivotality as the probabilistic object underlying pure-contribution incentives and characterises its non-monotonic relationship with background cooperation. Third, it shows that allowing probabilistic choices fundamentally changes the behavioural implications of universalisation by generating interior inclinations even when neither pure action is optimal. The same parameter σ captures both the counterfactual linkage between actions and the strength of the partial universalisation motive.

The rest of the paper is organised as follows. Section 2 introduces the threshold environment. Section 3 derives the collective-pivotality representation and establishes the strict single-peakedness of the contribution premium. Section 4 extends the analysis to mixed inclinations and studies the resulting choice structure. Section 5 discusses related literature, and Section 6 concludes.

2. MODEL ENVIRONMENT

Consider individuals interacting in groups of size $n \geq 3$. The public good is provided if at least $k \in \{2, \ldots, n-1\}$ members of the group contribute. Let $N = n-1$ denote the number of other group members. An individual takes an action $a_i \in \{0, 1\}$, where $a_i = 1$ denotes pure contribution and $a_i = 0$ denotes abstention. The material payoff of individual i is

$$\pi_i(a_i; a_{-i}) = B\mathbf{1}\left\{a_i + \sum_{j \neq i} a_j \geq k\right\} - ca_i,$$

where $B > 0$ is the benefit from public good provision and $c > 0$ is the cost of contribution. Define the cost-benefit ratio $\gamma = c/B$. We further take the background contribution probability $p \in [0, 1]$ as given.

We adopt the conditional-independence specification of Alger and Weibull (2016), under which the focal individual evaluates a contemplated strategy using a counterfactual profile in which each group member's strategy may be replaced by her own.

Assumption 1 (Conditional Independence). *Each of the N other group members independently adopts the focal individual's contemplated strategy with probability $\sigma \in [0, 1]$, and otherwise retains the background strategy. Background contribution decisions are mutually independent and independent of counterfactual replacement. Consequently, the number M of counterfactually linked members follows a binomial distribution, $M \sim \text{Bin}(N, \sigma)$.*

The parameter σ inherits its evolutionary interpretation from Alger and Weibull's *homo moralis* framework. In the decision problem studied here, however, it enters through the counterfactual replacement rule described above. Moreover, we call a group member *like-minded* if her strategy in the focal individual's counterfactual evaluation is replaced by the focal individual's contemplated strategy. A group member is *unlike-minded* if her strategy is not replaced and she instead retains the background contribution strategy. Finally, let

$$b_{m,j}(q) = \binom{m}{j} q^j (1-q)^{m-j}$$

denote the probability of exactly j contributions among m independent Bernoulli-q trials. The probability of at least t contributions is therefore

$$T_{m,t}(q) = \Pr[\text{Bin}(m, q) \geq t] = \sum_{j=t}^{m} b_{m,j}(q).$$

We adopt the conventions $T_{m,t}(q) = 1$ if $t \leq 0$, and $T_{m,t}(q) = 0$ if $t > m$.

3. PURE-STRATEGY BENCHMARK

We begin by restricting attention to the two pure strategies: to contribute or to abstain. When the focal individual contemplates an action, the like-minded members are assigned that same action. The question is therefore whether contribution is preferred to abstention once its consequences are evaluated collectively.

Under Assumption 1, each member is independently like-minded with probability σ, and unlike-minded with probability $1-\sigma$. Hence, the pure-commitment contribution probability of each member under contemplated action a is $q_a(p, \sigma) = \sigma a + (1-\sigma)p$. In particular, under contemplated contribution, $q_1 = \sigma + (1-\sigma)p$ and $q_0 = (1-\sigma)p$ is when the focal individual contemplates abstention. Her benefit-normalised counterfactual goal value is therefore

$$U(1; p) = T_{N,k-1}(\sigma + (1-\sigma)p) - \gamma$$

if she contributes, and

$$U(0; p) = T_{N,k}((1-\sigma)p)$$

if she does not. We call the contribution premium the collective-pivotality premium, indicating that the focal individual views her action as collectively pivotal. Formally, the premium is given by

$$\Delta^{PC}_{n,k}(p) = T_{N,k-1}(\sigma + (1-\sigma)p) - T_{N,k}((1-\sigma)p),$$

where PC denotes pure commitment. The focal individual prefers contribution to abstention if and only if $\Delta^{PC}_{n,k}(p) \geq \gamma$.

By interpreting the premium on a common probability space, we obtain Proposition 1.

Proposition 1 (Collective-pivotality representation). *Under common coupling, the premium is*

$$\Delta^{PC}_{n,k}(p) = \Pr(Q^1 \geq k > Q^0),$$

where Q^1 and Q^0 denote total contributions under the two counterfactuals, coupled using the same replace-

ment and background realisations.

Proof. See Appendix A. □

The representation in Proposition 1 gives the premium a direct probabilistic interpretation. Under unilateral reasoning, contribution is pivotal only when the focal individual's own action crosses the provision threshold. Under partial universalisation, the threshold may instead be crossed collectively by the focal individual and her like-minded members. This broader notion of pivotality suggests that the contribution premium is highest when background cooperation is sufficient to make provision attainable, but not so extensive that further contributions become redundant. The next subsection formalises this intuition.

3.1. MAIN RESULTS OF THE BENCHMARK

Theorem 1 (Single-peaked collective pivotality). *Suppose $n \geq 3$, $2 \leq k \leq n-1$, and $\sigma \in (0,1)$. There exists a unique $p^* \in (0,1)$ such that $\Delta^{PC}_{n,k}(p)$ is strictly single-peaked on $[0,1]$. Its unique peak is characterised implicitly by*

$$b_{n-2,k-2}\left(\sigma + (1-\sigma)p^*\right) = b_{n-2,k-1}\left((1-\sigma)p^*\right).$$

Proof Sketch. The proof of Theorem 1 uses the monotone likelihood ratio argument. It compares two adjacent binomial masses associated with the marginal threshold effects in the contribution- and abstention-oriented counterfactuals. Complete proof see Appendix A.

The left-hand side of the peak condition is the marginal probability that an increase in background cooperation crosses the threshold in the contribution-oriented counterfactual. The right-hand side is the marginal probability that the same increase causes the abstention-oriented counterfactual to reach the provision, making the contribution redundant. The peak occurs when these two effects exactly offset each other. A direct comparative static of the counterfactual linkage follows.

Proposition 2 (Counterfactual linkage). *For every $n \geq 3$, $2 \leq k \leq n-1$, and $p \in [0,1]$, the pure-contribution premium $\Delta^{PC}_{n,k}(p)$ is strictly increasing in σ. Consequently, for any fixed cost-benefit ratio γ, the contribution region expands with the strength of counterfactual linkage.*

Proof. See Appendix A. □

The proposition shows that stronger counterfactual linkage increases the probability of provision under contemplated contribution while decreasing it under contemplated abstention. The resulting increase in the contribution premium implies that the set of background contribution rates supporting pure contribution expands with σ. Whenever this set is bounded by two interior cutoffs, the lower cutoff decreases and the upper cutoff increases.

Next denote endpoint premiums to be:

$$\Delta_0 = \Delta^{PC}_{n,k}(0) = T_{N,k-1}(\sigma),$$

and

$$\Delta_1 = \Delta^{PC}_{n,k}(1) = 1 - T_{N,k}(1-\sigma).$$

The two expressions tell us the behaviour at two ends of baseline contribution probabilities. We impose the following assumption such that with the cost-benefit ratio above both endpoint premiums but below the unique maximum. Together with strict single-peakedness, this guarantees two interior cutoffs and a non-degenerate contribution region.

Assumption 2 (Regular intermediate-cost environment). *Assume the cost-benefit ratio satisfies*

$$\max\{\Delta_0, \Delta_1\} < \gamma < \bar{\Delta},$$

where $\bar{\Delta} = \Delta^{PC}_{n,k}(p^*)$ denote the maximum premium.

Proposition 3 (Bounded pure-contribution preference interval). *Under Assumption 2, there exist unique cutoffs $0 < p_- < p^* < p_+ < 1$ such that $\Delta^{PC}_{n,k}(p_-) = \Delta^{PC}_{n,k}(p_+) = \gamma$. The pure-commitment ranking*

is then

$$1 \succ 0 \iff p \in (p_-, p_+),$$
$$1 \sim 0 \iff p \in \{p_-, p_+\},$$
$$1 \prec 0 \iff p \in [0, p_-) \cup (p_+, 1].$$

Proof. See Appendix A. □

The three regions reflect the competing effects of background cooperation on collective pivotality. At low background contribution rates, increasing p raises the contribution premium by making threshold crossing under the contribution-oriented counterfactual more likely. At high rates, further increases in p reduce the premium as provision under the abstention-oriented counterfactual becomes relatively more likely. Under Assumption 2, these opposing effects generate a bounded interval over which contribution is strictly preferred, with abstention preferred on either side. Thus, identical abstention behaviour can arise on opposite sides of the collective-pivotality peak, reflecting different responses to background cooperation.

Before ending this section, we introduce a simplified example to illustrate the mechanism.

Example . *Consider $n = 3$ and $k = 2$. If at least one partner is like-minded, the focal individual's action is collectively pivotal regardless of the remaining background action. Collective pivotality can therefore fail only when no partner is linked, in which case the two background actions must be either both zero or both one. Hence*

$$\Delta^{\mathrm{PC}}_{3,2}(p) = 1 - (1-\sigma)^2\left[p^2 + (1-p)^2\right].$$

The premium is symmetric and uniquely maximised at $p^* = 1/2$. Thus, for

$$(2-\sigma)\sigma < \gamma < 1 - \frac{(1-\sigma)^2}{2},$$

contribution is strictly preferred on an open interval around $1/2$.

4. MIXED STRATEGY AND UNIVERSALISED INCLINATIONS

We now extend the choice set from pure commitments to contribution inclinations $a \in [0, 1]$. An inclination specifies the probability with which the focal individual contributes. Under partial universalisation, like-minded members adopt the same inclination, whereas unlike-minded members retain the background contribution probability p. Conditional on these strategies, actions are independently realised. Thus, the contribution probability of each co-player is

$$q(a, p) = \sigma a + (1-\sigma)p.$$

The focal individual's objective is then

$$U(a; p) = T_{N,k}(q(a, p)) + a b_{N,k-1}(q(a, p)) - \gamma a.$$

Since $U(a; p)$ is continuous on $[0, 1]$, the best-response set

$$BR(p) = \arg\max_{a \in [0,1]} U(a; p)$$

is non-empty and compact. Unlike ordinary mixed-strategy evaluation, changing a also changes the counterfactual contribution probabilities of like-minded members. The objective is therefore generally non-linear in a, allowing a strictly optimal interior inclination rather than mixing solely through indifference.

We now examine conditions under which partial universalisation generates a unique interior contribution inclination. For general (n, k), the objective need not be globally concave, so first-order conditions alone may not identify the global maximiser. We therefore establish sufficient conditions for a unique interior optimum

and prove that they hold on a non-empty open parameter set for every admissible threshold. First, define

$$m_0(\sigma, p) = b_{N,k-1}(q_0) + \sigma N b_{N-1,k-1}(q_0),$$

$$m_1(\sigma, p) = b_{N,k-1}(q_1) + \sigma N b_{N-1,k-2}(q_1).$$

These are the gross marginal benefits of increasing the contribution inclination at the two endpoints. Then, we claim that the sufficient region that exists under our construction is

$$\mathcal{O}_{n,k} = \left\{ (\sigma, p, \gamma) \in (0,1)^2 \times (0,\infty) : \frac{\partial^2 U(a;p)}{\partial a^2} < 0 \text{ for all } a \in [0,1], \quad m_1(\sigma,p) < \gamma < m_0(\sigma,p) \right\}.$$

Theorem 2 (Interior inclination with robustness). *For every $n \geq 3$ and $2 \leq k \leq n-1$, the set $\mathcal{O}_{n,k}$ is non-empty and open. For every $(\sigma, p, \gamma) \in \mathcal{O}_{n,k}$, the mixed-strategy problem has a unique global maximiser $a^*(\sigma, p, \gamma) \in (0,1)$.*

Proof. See Appendix A. □

Theorem 2 identifies a form of strict interior choice. On $\mathcal{O}_{n,k}$, neither certain abstention nor certain contribution is optimal. The focal individual instead strictly prefers an intermediate contribution inclination. In particular, at $a = 0$, the benefit from slightly increasing the contribution inclination exceeds its cost, whereas at $a = 1$ the marginal benefit falls below cost. On this parameter region, the marginal benefit of increasing the contribution inclination declines, producing a unique interior optimum where it equals the marginal cost. An intermediate inclination balances the direct cost of contribution against this changing collective effect. Moreover, non-emptiness shows that this mechanism is available for every admissible threshold (n, k), while openness gives it local robustness. In particular, the theorem survives small changes in the counterfactual linkage strength σ, the background contribution rate, p, or the cost γ. Such interior optima are therefore not knife-edge.

The following corollary relates the interior optimum to the ranking of pure commitments.

Corollary 1 (Abstention-inclination reversal). *If $(\sigma, p, \gamma) \in \mathcal{O}_{n,k}$ and $\gamma > \Delta^{\mathrm{PC}}_{n,k}(p)$, then the pure-strategy problem strictly selects abstention, while the mixed-strategy problem has a unique interior optimum $a^* \in (0,1)$, with*

$$U(a^*; p) > U(0; p) > U(1; p).$$

The result follows immediately from the interior-optimum theorem and the definition of the pure-contribution premium. The reversal shown by Corollary 1 clarifies what pure abstention means. A preference for abstention over certain contribution does not imply that the individual prefers zero contribution inclination. It may instead reflect that full contribution is too strong a response. Once intermediate inclinations are available, the individual can strictly prefer an intermediate response. Conversely, if pure contribution is preferred in a pure commitment environment, a lower intermediate inclination may still be better than it. Thus, the mixed-strategy problem refines the binary question of whether to contribute to the continuous question of how strongly to do so.

Theorem 2 gives a general sufficient condition for a unique interior optimum, but global concavity is not necessary for such behaviour. Without it, a complete solution for arbitrary (n, k) is less tractable. The three-player case below however is simple enough to solve globally and shows that the interior result extends beyond the concave region.

Example . *Again let $n = 3$ and $k = 2$. Define*

$$A_0(\sigma, p) = 2q_0(1 + \sigma - q_0), \qquad A_1(\sigma, p) = 2(1 - q_1)(q_1 + \sigma),$$

and

$$E(\sigma, p) = 2 + \sigma - 4q_0.$$

Whenever

$$A_1(\sigma, p) < \gamma < A_0(\sigma, p),$$

the mixed-strategy problem has a unique global maximiser

$$a^*(\sigma, p, \gamma) = \frac{E + \sqrt{E^2 + 6(A_0 - \gamma)}}{6\sigma} \in (0, 1).$$

Moreover,

$$A_0(\sigma, p) - A_1(\sigma, p) = 4\sigma(1 - \sigma)(2p - 1),$$

so this region is non-empty whenever $p > 1/2$.

The example shows that strict global concavity is sufficient rather than necessary. In the more tractable three-player case, the endpoint marginal condition alone delivers a unique interior global optimum over a larger parameter region.

4.1. TWO BOUNDARY CASES AND COMPARATIVE STATICS

We now examine the behavioural implications of the optimal contribution inclination $a^*(\sigma, p, \gamma)$. We first contrast partial universalisation with the two boundary cases, *homo oeconomicus* and *homo kantiensis*, before studying comparative statics with respect to counterfactual linkage, contribution costs, and background cooperation.

The two boundary cases isolate the two forces that are combined under partial universalisation. When $\sigma = 0$, the focal individual evaluates her mixed strategy while holding the behaviour of others fixed, so $q(a, p) = p$. Hence, the objective reduces to

$$U_{\sigma=0}(a; p) = T_{N,k}(p) + a[b_{N,k-1}(p) - \gamma].$$

The background contribution rate p affects which pure action is preferred, but an interior inclination can therefore be optimal only through indifference, and is never uniquely optimal. At the opposite boundary, $\sigma = 1$, all members use the same contribution inclination as the focal individual. The background contribution then disappears from the problem,

$$U_{\sigma=1}(a) = [T_{N,k}(a) + a b_{N,k-1}(a)] - \gamma a = T_{n,k}(a) - \gamma a.$$

Full universalisation may therefore support an interior inclination, but that inclination cannot respond to the surrounding level of contribution p. Partial universalisation can combine these two features. For $0 < \sigma < 1$, the objective may have a strict interior optimum while the optimal inclination remains responsive to the social background. This distinguishes the partial regime from both boundary cases and motivates the comparative statics below.

Throughout, subscripts on U denote partial derivatives. Unlike the pure-contribution premium in Proposition 2, the optimal inclination need not increase with counterfactual linkage.

Proposition 4 (Comparative static of the counterfactual linkage). *Suppose $(\sigma, p, \gamma) \in \mathcal{O}_{n,k}$, and let $q^* = p + \sigma(a^* - p)$. Define*

$$\Lambda(q, p) = (n - 1)\left[(1 - p) b_{n-2,k-1}(q) + p\, b_{n-2,k-2}(q)\right].$$

Then $\Lambda(q^*, p) > 0$ and

$$\frac{\partial a^*}{\partial \sigma} = \frac{\Lambda(q^*, p)}{-U_{aa}(a^*; p)} - \frac{a^* - p}{\sigma}.$$

Consequently, $\partial a^*/\partial \sigma > 0$ whenever $a^* \leq p$. When $a^* > p$, either sign is possible; both signs occur on non-empty open subsets of $\mathcal{O}_{3,2}$.

Proof. See Appendix A. □

Unlike the pure-contribution premium, the optimal inclination need not increase with counterfactual linkage. Stronger linkage makes changes in the focal inclination more consequential for the counterfactual group, but also shifts the group's contribution probabilities away from the background rate towards the focal

inclination. These competing effects imply that a stronger linkage may reduce the optimal inclination when it already exceeds background cooperation.

The response to changes in contribution cost is straightforward. Since $U_{a\gamma}(a; p, \gamma) = -1$, differentiating the first-order condition gives

$$\frac{\partial a^*}{\partial \gamma} = -\frac{U_{a\gamma}(a^*; p, \gamma)}{U_{aa}(a^*; p, \gamma)} = \frac{1}{U_{aa}(a^*; p, \gamma)} < 0.$$

A higher contribution cost therefore reduces the optimal contribution inclination throughout the interior region. The response to changes in background contribution rates is less mechanical. Differentiating with respect to p gives

$$\frac{\partial a^*}{\partial p} = -\frac{U_{ap}(a^*; p, \gamma)}{U_{aa}(a^*; p, \gamma)}.$$

Because a change in p affects the probability of provision at every contemplated inclination, the sign of this expression is not determined by the second-order condition alone. Nevertheless, the same local geometry that supports a robust interior solution also generates a robust region in which greater background participation lowers the optimal inclination.

Proposition 5 (Comparative statics of the interior inclination). *For every admissible threshold (n, k), the optimal interior inclination $a^* : \mathcal{O}_{n,k} \to (0, 1)$ is continuously differentiable. Moreover, there exists a non-empty open subset $\mathcal{R}_{n,k} \subseteq \mathcal{O}_{n,k}$ such that*

$$\frac{\partial a^*}{\partial \gamma} < 0 \quad \text{on } \mathcal{O}_{n,k}, \qquad \frac{\partial a^*}{\partial p} < 0 \quad \text{on } \mathcal{R}_{n,k}.$$

Proof. See Appendix A. □

The background effect is less immediate than the cost effect. On $\mathcal{R}_{n,k}$, greater background cooperation reduces the marginal benefit of increasing the focal inclination, leading to a lower optimal contribution probability. This crowding-out effect is local: it does not imply that the optimal inclination must vanish as background cooperation approaches certainty. The following example shows that a strictly positive interior inclination can persist even in highly cooperative environments.

Example . *Let $n = 3$, $k = 2$, and suppose*

$$0 < \sigma < \frac{2}{3}.$$

If

$$\sigma(2 - \sigma) < \gamma < 4\sigma(1 - \sigma),$$

then there exists $\bar{p} < 1$ such that, for every $p \in (\bar{p}, 1]$,

1. *the pure-strategy problem under partial universalisation strictly prefers abstention;*
2. *the mixed-strategy problem under partial universalisation has a unique optimum*

$$a^*(\sigma, p, \gamma) \in (0, 1);$$

3. *ordinary expected-payoff reasoning, $\sigma = 0$, strictly prefers abstention.*

The example separates inclination from simple free-riding. As p becomes large, the optimal interior inclination may decline, but it need not disappear. Indeed, there are environments in which both ordinary reasoning and the pure-strategy comparison under partial universalisation select abstention, while allowing the contribution inclination to vary still yields a strictly positive interior response. Partial universalisation can therefore sustain inclined participation even when the social background is already highly supportive.

Taken together, these results clarify what is distinctive about the partial regime. Ordinary reasoning allows behaviour to respond to the background contribution but generically selects an endpoint. Full universalisation may generate an interior response but removes the background contribution from the decision problem. Partial universalisation can instead support an interior inclination that adjusts continuously

to both private incentives and the background participation rate, while remaining positive even in some high-support environments.

5. RELATED LITERATURE

The collective-pivotality results established in Section 3 bring together two strands of research that have largely developed separately. The classical literature on finite threshold public goods studies how participation depends on an individual's probability of being pivotal for provision. In the no-refund game of Palfrey and Rosenthal (1984), a unilateral contribution changes the outcome precisely when $k-1$ of the other $n-1$ players contribute. Accordingly, at $\sigma = 0$, the premium reduces to the familiar binomial pivotal probability $b_{n-1,k-1}(p)$. Allowing the contemplated action to be partially inherited by other group members expands this event from unilateral to collective pivotality: provision must lie between two coupled counterfactual profiles generated by a random block of linked members. The strict single-peakedness of $\Delta^{PC}_{n,k}(p)$ shows that the pivotality logic of the classical model survives this transformation, while its pointwise increase in σ describes how stronger counterfactual linkage expands the contribution region. Related work introduces moral considerations into threshold participation through heterogeneous costs of shared and pivotal guilt, as in Rothenhäusler, Schweizer, and Szech (2018). Together, these approaches illustrate how threshold aggregation connects material pivotality, perceived responsibility, and morally motivated participation.

The counterfactual linkage used here comes from the theory of *homo moralis*. Alger and Weibull (2013) derive the preference in pairwise interactions, and Alger and Weibull (2016) extend the evolutionary analysis to groups of arbitrary finite size. Under conditional independence, the stable morality profile has a binomial representation in which each co-player's strategy is independently replaced by the focal individual's contemplated strategy with a common probability. [1] Complementary axiomatic work studies which preferences admit representations based on universalised counterfactuals (Salonia , 2024). These contributions provide the preference foundation taken as given in the present analysis; the question pursued above is what behaviour that preference induces once material payoffs are aggregated through a discontinuous provision rule.

Related applications of probabilistic universalisation arise in voting. In the voting model of Alger and Laslier (2022), a juror is κ-pivotal for a deviation when it changes the collective decision in a counterfactual scenario in which at least one other juror also adopts the deviating strategy, each doing so independently with probability κ. Their analysis shows how this form of pivotality shapes coordination and information aggregation in Bayesian voting equilibria. Alger, Dierks, and Laslier (2026) subsequently use universalisation to study costly turnout in continuum electorates with core constituencies and power-sharing electoral outcomes. These applications suggest that probabilistic universalisation is especially consequential when individual participation interacts with an aggregate decision rule. The threshold-straddling representation in this paper isolates that interaction in a finite public-good environment and reveals its dependence on the surrounding contribution rate. Here, at low background cooperation rates, further participation raises the contribution premium by facilitating threshold crossing; at high rates, it reduces the premium as provision under abstention becomes more likely.

A related literature studies how *homo moralis* affects public-good provision and coordination. Alger and Weibull (2017) show how morality changes strategic behaviour in continuous-contribution public goods and coordination games. Allowing morality to vary across agents, Eichner and Runkel (2024) demonstrate that strategic substitution can generate contribution polarization and can cause less moral agents to offset the provision of more moral agents. Alger (2026) combines a fully Kantian personal norm with inequity aversion to account for unconditional and conditional cooperation, and extends the analysis to threshold public goods. Under complete universalisation, the exact threshold drops out of the Kantian comparison between fully universalised pure profiles. The $\sigma = 1$ boundary recovers this invariance for the two deterministic commitments while also showing why it need not extend to the full inclination problem. Partial universalisation makes probability mass near the threshold central, whereas the objective over mixed inclinations can retain threshold dependence even when the endpoint comparison does not.

Universalisation also bears on the choice of mixed actions. Alger and Weibull (2013) already show that *homo moralis* utility can be non-linear in one's own mixing probability and can select an interior

[1] Alger and Weibull (2023) revise one instability claim in the n-player analysis; the stable morality profile and the conditional-independence used here are unaffected.

mixture. More recently, Suresh and Studtmann (2026) obtained a symmetric equilibrium in which agents randomise between selfish withholding and fully Kantian contribution in a linear public-good game. Their analysis shows how equilibrium indifference between composite morality-action strategies can sustain interior participation. Our inclination results identify a complementary channel. Under a fixed degree of partial universalisation and a fixed background environment, an interior inclination can be strictly preferred to both deterministic commitments. For every admissible threshold, such strict interiority arises on a non-empty open set of parameters; under the additional reversal conditions, it can occur even when $\Delta^{PC}_{n,k}(p) < \gamma$, so that abstention is preferred in the comparison of the two endpoints. Threshold aggregation can therefore separate the ranking of deterministic commitments from the ranking over the full inclination simplex.

The broader literature offers several related ways of formalising Kantian or action-linked reasoning. The Kantian-economics tradition is initiated by Laffont (1975) and the Kantian equilibrium that studies common proportional or additive deviations evaluated through agents' individual payoffs are in Roemer (2010, 2015), while Kordonis (2020) lets an individual evaluate a common strategy rule through a risk-sensitive welfare criterion over positions in an imagined virtual group. Action linkage can instead enter beliefs. In Masel (2007) and Daley and Sadowski (2017), the individual's contemplated action changes the distribution she assigns to others' actual behaviour. Taken together, these approaches locate universalisation in different components of the decision problem. Work on the evolutionary stability of multidimensional moral foundations provides a broader account in which several moral principles may coexist (Avataneo, Norman, and Persico, 2025); fixing the scalar *homo moralis* profile allows the threshold mechanism itself to be characterised separately.

Finally, empirical research motivates the ingredients of this mechanism and suggests how its implications might be tested. Universalisation predicts moral-permissibility judgments in threshold problems (Levine, et al. , 2020), while structural estimates from pairwise laboratory games indicate heterogeneous and non-degenerate Kantian concern alongside distributional preferences and reciprocity (van Leeuwen and Alger , 2024). Broader cross-country evidence likewise shows that cooperative choices jointly reflect beliefs, perceived norms, and preferences (Andre, et al. , 2026). These findings motivate allowing both universalisation concern and the surrounding rate of cooperation to enter the decision problem. They also leave open direct tests of the collective-pivotality channel: whether contribution is highest at intermediate background cooperation, whether stronger perceived linkage expands the contribution region, and whether deterministic choices conceal a preferred interior inclination.

6. CONCLUSION

Understanding collective action requires not only identifying which motivational dispositions can survive evolutionary selection, but also characterising the behaviour they induce in particular decision environments. This paper characterises how partial universalisation shapes contribution decisions in finite threshold public goods. For pure commitments, the contribution premium admits a collective-pivotality representation: it equals the probability that contribution- and abstention-oriented counterfactuals lie on opposite sides of the provision threshold. For every $0 < \sigma < 1$, this premium is strictly single-peaked in background cooperation, generating a bounded contribution region under intermediate cost-benefit ratios. Extending the choice set to contribution inclinations, we establish that every admissible group size and threshold admits a non-empty open set of parameters supporting a unique interior optimum. Moreover, such an optimum can coexist with a strict preference for pure abstention over pure contribution. The ranking of deterministic commitments therefore need not determine the optimal contribution inclination.

The comparison with the two boundary cases clarifies what distinguishes partial universalisation. Under ordinary reasoning, $\sigma = 0$, the objective is affine in the contribution inclination and therefore generically selects an endpoint. Under complete universalisation, $\sigma = 1$, an interior inclination may be optimal, but background cooperation disappears from the decision problem. Partial universalisation can instead support a strict interior optimum that responds to the surrounding level of cooperation. Furthermore, stronger counterfactual linkage always raises the pure-contribution premium but need not increase the optimal inclination. Greater background cooperation can also locally crowd out the inclination, although a positive interior optimum may persist even when background participation approaches certainty. These results are established while taking background cooperation as exogenous and providing sufficient rather than exhaustive conditions for interior choice. A natural extension is to characterise the full best-response correspondence and equilibria with endogenous background cooperation.

The results also have implications for interpreting observed participation. Abstention may arise at both low and high levels of background cooperation, where the relative importance of threshold activation and redundancy differs. Moreover, an observed abstention may be the realisation of a strictly positive optimal contribution inclination, even when certain abstention is preferred to certain contribution. Binary participation therefore need not reveal the underlying ranking of contribution strategies. The distinction between collective pivotality and optimal inclinations suggests empirical tests of how contribution responds to background cooperation and counterfactual linkage, while highlighting the limitations of inferring cooperative dispositions from observed actions alone.

A. APPENDIX

Lemma 1 For $m \geq 1$ and $t \in \{1, \ldots, m\}$,

$$\frac{\partial}{\partial q} T_{m,t}(q) = m b_{m-1,t-1}(q).$$

Proof. By definition,

$$T_{m,t}(q) = \sum_{j=t}^{m} \binom{m}{j} q^j (1-q)^{m-j}.$$

Differentiating term by term gives

$$\frac{\partial}{\partial q} T_{m,t}(q) = \left(\sum_{j=t}^{m} j \binom{m}{j} q^{j-1} (1-q)^{m-j} \right) - \left(\sum_{j=t}^{m} (m-j) \binom{m}{j} q^j (1-q)^{m-j-1} \right).$$

Using the binomial identities

$$j \binom{m}{j} = m \binom{m-1}{j-1}$$

and

$$(m-j) \binom{m}{j} = m \binom{m-1}{j},$$

we obtain

$$\frac{\partial}{\partial q} T_{m,t}(q) = \left(m \sum_{j=t}^{m} \binom{m-1}{j-1} q^{j-1} (1-q)^{m-j} \right) - \left(m \sum_{j=t}^{m-1} \binom{m-1}{j} q^j (1-q)^{m-j-1} \right).$$

Changing the index in the first sum from j to $r = j-1$,

$$\frac{\partial}{\partial q} T_{m,t}(q) = \left(m \sum_{r=t-1}^{m-1} \binom{m-1}{r} q^r (1-q)^{m-1-r} \right) - \left(m \sum_{r=t}^{m-1} \binom{m-1}{r} q^r (1-q)^{m-1-r} \right).$$

All terms cancel except the term with $r = t-1$. Hence

$$\frac{\partial}{\partial q} T_{m,t}(q) = m \binom{m-1}{t-1} q^{t-1} (1-q)^{m-t},$$

which is $m b_{m-1,t-1}(q)$. □

Lemma 2 For $m \geq 1$ and $t \in \{1, \ldots, m\}$,

$$\frac{\partial}{\partial a} b_{m,t}(\sigma a + (1-\sigma)p) = \sigma m \left[b_{m-1,t-1}(\sigma a + (1-\sigma)p) - b_{m-1,t}(\sigma a + (1-\sigma)p) \right],$$

and

$$\frac{\partial}{\partial a} T_{m,t}(\sigma a + (1-\sigma)p) = \sigma m b_{m-1,t-1}(\sigma a + (1-\sigma)p).$$

Proof. By definition,

$$b_{m,t}(\sigma a + (1-\sigma)p) = \binom{m}{t} (\sigma a + (1-\sigma)p)^t (1 - \sigma a - (1-\sigma)p)^{m-t}.$$

Differentiate with respect to a,

$$\begin{aligned}\frac{\partial}{\partial a}b_{m,t} &= t\binom{m}{t}\sigma(\sigma a+(1-\sigma)p)^{t-1}(1-\sigma a-(1-\sigma)p)^{m-t}- \\ &\quad (m-t)\binom{m}{t}\sigma(\sigma a+(1-\sigma)p)^{t}(1-\sigma a-(1-\sigma)p)^{m-t-1} \\ &= m\binom{m-1}{t-1}\sigma(\sigma a+(1-\sigma)p)^{t-1}(1-\sigma a-(1-\sigma)p)^{m-t}- \\ &\quad m\binom{m-1}{t}\sigma(\sigma a+(1-\sigma)p)^{t}(1-\sigma a-(1-\sigma)p)^{m-t-1} \\ &= \sigma m\left[b_{m-1,t-1}(\sigma a+(1-\sigma)p)-b_{m-1,t}(\sigma a+(1-\sigma)p)\right],\end{aligned}$$

as required. For the second statement, by definition,

$$T_{m,t}(\sigma a+(1-\sigma)p)=\sum_{j=t}^{m}\binom{m}{j}(\sigma a+(1-\sigma)p)^{j}(1-\sigma a-(1-\sigma)p)^{m-j}.$$

Differentiating term by term gives,

$$\begin{aligned}\frac{\partial}{\partial a}T_{m,t}(\sigma a+(1-\sigma)p) &= \left(\sum_{j=t}^{m}j\binom{m}{j}\sigma(\sigma a+(1-\sigma)p)^{j-1}(1-\sigma a-(1-\sigma)p)^{m-j}\right)- \\ &\quad \left(\sum_{j=t}^{m}(m-j)\binom{m}{j}\sigma(\sigma a+(1-\sigma)p)^{j}(1-\sigma a-(1-\sigma)p)^{m-j-1}\right) \\ &= \sigma m\left(\sum_{j=t}^{m}b_{m-1,j-1}(\sigma a+(1-\sigma)p)-b_{m-1,j}(\sigma a+(1-\sigma)p)\right) \\ &= \sigma m b_{m-1,t-1}(\sigma a+(1-\sigma)p),\end{aligned}$$

as required. □

Proof of Proposition 1. To interpret the premium on a common probability space, we apply common coupling. For each partner $j=1,\ldots,N$, let $S_j\sim\text{Bernoulli}(\sigma)$ indicate whether that member is like-minded, and let $Z_j\sim\text{Bernoulli}(p)$ denote the unlike-minded member's baseline action. Assume that all S_j and Z_j are mutually independent. Then, the member's action distribution under focal action a is $Y_j^a=S_ja+(1-S_j)Z_j$. Define $M=\sum_{j=1}^{N}S_j$ and $R=\sum_{j=1}^{N}(1-S_j)Z_j$, where M is the number of members that are like-minded, while R is the number of baseline contributors among the unlike-minded members. Total contribution under the two contemplated actions are:

$$Q^1=1+\sum_{j=1}^{N}Y_j^1=1+M+R,$$

and

$$Q^0=\sum_{j=1}^{N}Y_j^0=R.$$

Since $Q^1\geq Q^0$ for every realisation, $\{Q^0\geq k\}\subseteq\{Q^1\geq k\}$. Therefore,

$$\begin{aligned}\Delta_{n,k}^{PC}(p) &= \Pr(Q^1\geq k)-\Pr(Q^0\geq k) \\ &= \Pr(Q^1\geq k>Q^0).\end{aligned}$$

Substituting $Q^1 = 1 + M + R$ and $Q^0 = R$, the straddling event becomes

$$1 + M + R \geq k \qquad \text{and} \qquad R < k.$$

Because R is an integer, this is equivalent to

$$R \leq k - 1 \leq R + M.$$

Finally, by Assumption 1, $M \sim \text{Bin}(N, \sigma)$. Conditional on $M = m$, exactly $N - m$ partners retain their baseline actions, each independently contributing with probability p. Hence

$$R \mid M = m \sim \text{Bin}(N - m, p).$$

The event

$$R \leq k - 1 \leq R + m$$

is equivalent to

$$k - 1 - m \leq R \leq k - 1.$$

Accounting for the lower boundary $R \geq 0$ and summing over m completes the proof. □

Proof of Theorem 1. For notational convenience, write $\Delta(p) = \Delta^{PC}_{n,k}(p)$. Define $q_1 = \sigma + (1 - \sigma)p$ and $q_0 = (1 - \sigma)p$. Then

$$\Delta(p) = T_{N,k-1}(q_1) - T_{N,k}(q_0).$$

Applying Lemma 1 and the chain rule,

$$\begin{aligned}\Delta'(p) &= N(1 - \sigma)b_{N-1,k-2}(q_1) - N(1 - \sigma)b_{N-1,k-1}(q_0). \\ &= N(1 - \sigma)[b_{N-1,k-2}(q_1) - b_{N-1,k-1}(q_0)].\end{aligned}$$

Let $m = N - 1 = n - 2$ and $z = (1 - \sigma)p$. Then,

$$z \in [0, 1 - \sigma], \qquad q_0 = z, \qquad q_1 = z + \sigma.$$

For $z \in (0, 1 - \sigma)$, define the likelihood ratio

$$R(z) = \frac{b_{m,k-2}(z + \sigma)}{b_{m,k-1}(z)}.$$

Using the binomial mass formula,

$$R(z) = \frac{k - 1}{n - k} \times \frac{(z + \sigma)^{k-2}(1 - z - \sigma)^{n-k}}{z^{k-1}(1 - z)^{n-k-1}}.$$

Its logarithmic derivative is

$$\begin{aligned}\frac{\partial}{\partial z} \log R(z) &= \frac{k - 2}{z + \sigma} - \frac{k - 1}{z} + \frac{n - k - 1}{1 - z} - \frac{n - k}{1 - z - \sigma} \\ &= -\frac{z + (k - 1)\sigma}{z(z + \sigma)} - \frac{(1 - z) + (n - k - 1)\sigma}{(1 - z)(1 - z - \sigma)} < 0.\end{aligned}$$

Therefore,

$$\frac{\partial}{\partial z} \log R(z) < 0,$$

$R(z)$ is strictly decreasing. As $z \to 0^+$, $b_{m,k-1}(z) \to 0$, whereas $b_{m,k-2}(z + \sigma) > 0$. Hence $R(z) \to +\infty$. As $z \to (1 - \sigma)^-$, $(1 - z - \sigma) \to 0^+$. As $z \to (1 - \sigma)^-$, $b_{m,k-2}(z + \sigma) \to 0$, because $z + \sigma \to 1$ and $k - 2 < m$. Meanwhile, $b_{m,k-1}(z) \to b_{m,k-1}(1 - \sigma) > 0$. Hence $R(z) \to 0$. By continuity and strict monotonicity, there

exists a unique $p^* \in (0,1)$ such that $R(z^*) = 1$. Equivalently,

$$b_{m,k-2}(z^* + \sigma) = b_{m,k-1}(z^*).$$

Define

$$p^* = \frac{z^*}{1-\sigma}.$$

Then $p^* \in (0,1)$. Since the sign of $\Delta'(p)$ is the sign of $R((1-\sigma)p) - 1$, because R is strictly decreasing, this expression is positive below p^* and negative above p^*. This proves strict single-peakedness. □

Proof of Proposition 2. Let $q_1 = \sigma + (1-\sigma)p$ and $q_0 = (1-\sigma)p$. By Lemma 1,

$$\frac{\partial}{\partial \sigma}\Delta^{PC}_{n,k}(p) = (n-1)\left[(1-p)b_{n-2,k-2}(q_1) + pb_{n-2,k-1}(q_0)\right] > 0.$$

For $p \in (0,1)$, both terms are positive; at $p = 0$ and $p = 1$, respectively, the first and second term is positive. Hence $\Delta^{PC}_{n,k}(p)$ is strictly increasing in σ. Therefore, if $\sigma_2 > \sigma_1$,

$$\{p : \Delta^{PC}_{n,k,\sigma_1}(p) \geq \gamma\} \subseteq \{p : \Delta^{PC}_{n,k,\sigma_2}(p) \geq \gamma\},$$

which proves the second claim. □

Proof of Proposition 3. By Theorem 1, $\Delta(p)$ is continuous and strictly increasing on $[0, p^*]$, then strictly decreasing on $[p^*, 1]$. Assumption 2 implies $\Delta(0) < \gamma < \Delta(p^*)$ and $\Delta(1) < \gamma < \Delta(p^*)$. By continuity and strict monotonicity on the increasing branch, there exists a unique $p_- \in (0, p^*)$ such that $\Delta(p_-) = \gamma$ Similarly, continuity and strict monotonicity on the decreasing branch imply that there exists a unique $p_+ \in (p^*, 1)$ such that $\Delta(p_+) = \gamma$. The sign pattern is therefore

$$\Delta(p) < \gamma \qquad \text{for } p < p_-, \Delta(p) \geq \gamma \qquad \text{for } p \in [p_-, p_+],$$

and

$$\Delta(p) < \gamma \qquad \text{for } p > p_+.$$

Under the contribution condition $\Delta^{PC}_{n,k}(p) \geq \gamma$, this yields the pure-commitment ranking in Proposition 3. □

Proof of Theorem 2. Step 1: Define a sufficient optimisation region. Let $\theta = (\sigma, p, \gamma)$ and $\mathcal{D} = (0,1) \times (0,1) \times (0,\infty)$. Recall that we have defined

$$m_0(\sigma, p) = b_{N,k-1}(q_0) + \sigma N b_{N-1,k-1}(q_0) = U'(0; p) + \gamma,$$

and

$$m_1(\sigma, p) = b_{N,k-1}(q_1) + \sigma N b_{N-1,k-2}(q_1) = U'(1; p) + \gamma.$$

We claim that $m_0(\sigma, p) > m_1(\sigma, p)$ under the construction, as will be proved in *Step 2*. We need to show that for every admissible threshold of the following set, it is non-empty and open:

$$\mathcal{O}_{n,k} = \left\{\theta \in \mathcal{D} : \max_{a \in [0,1]} U''(a; \theta) < 0, \quad m_1(\sigma, p) < \gamma < m_0(\sigma, p)\right\}.$$

We use $U''(\cdot)$ to represent the second derivative with respect to a, and similarly, $U'(\cdot)$ will be the first-order derivative with respect to a. At any parameter vector in this set, U' is strictly decreasing on $[0,1]$, positive at zero, and negative at one. Hence we can obtain a unique global maximiser. We need to show that such parameters exist for every admissible threshold and that $\mathcal{O}_{n,k}$ is open.

Step 2: Construct a non-empty parameter region. Recall that the focal individual's objective is

$$U(a; p) = T_{N,k}(q(a,p)) + ab_{N,k-1}(q(a,p)) - \gamma a.$$

Write $A(q) = T_{N,k}(q)$ and $C(q) = b_{N,k-1}(q)$, where $q = q(a, p)$. The objective function becomes

$$U(a; p) = A(q) + aC(q) - \gamma a.$$

Therefore,

$$U'(a; p) = \sigma A'(q) + C(q) + a\sigma C'(q) - \gamma,$$

and

$$U''(a; p) = 2\sigma C'(q) + \sigma^2[A''(q) + aC''(q)].$$

For each admissible (n, k), we seek parameters (p, σ) such that $U''(a; p) < 0$ for all $a \in [0, 1]$. Since the curvature vanishes at zero linkage ($\sigma = 0$), continuity at that boundary alone cannot determine its sign for positive linkage. Hence, we first control the term linear in σ, and then uniformly bound the remaining terms. For $q \in (0, 1)$, by product rule,

$$C'(q) = C(q)\frac{k - 1 - Nq}{q(1 - q)}.$$

Choose $p_0 \in ((k - 1)/N, 1)$. This interval is non-empty because $k \leq N$, and at p_0, $C'(p_0) < 0$. Now by continuity of $C'(q)$, choose $\varepsilon, \eta > 0$ such that,

$$\mathcal{J} = [p_0 - \varepsilon, p_0 + \varepsilon] \subset (0, 1), \qquad C'(q) \leq -\eta, \forall q \in \mathcal{J},$$

where $\mathcal{J}$ is compact. Next we attempt to uniformly bound the term that is quadratic in σ. Because A and C are polynomials, their derivatives are continuous, and the function $(q, a) \mapsto |A''(q) + aC''(q)|$ is continuous. It therefore attains a finite maximum $K \geq 0$ on the compact set $\mathcal{J} \times [0, 1]$, formally,

$$|A''(q) + aC''(q)| \leq K \qquad \forall (q, a) \in \mathcal{J} \times [0, 1].$$

Moreover, at $p = p_0$, we have $q = p_0 + \sigma(a - p_0)$, and $\sup_{a\in[0,1]} |q - p_0| \leq \sigma$. Thus, if we choose $0 < \sigma < \min\{1, \varepsilon, \eta/(K + 1)\}$, then every contemplated inclination generates $q \in \mathcal{J}$, and

$$U''(a; p) \leq -2\eta\sigma + K\sigma^2 < -\eta\sigma < 0 \qquad \forall a \in [0, 1].$$

This establishes strict concavity uniformly across the entire choice interval. It suffices to show that the unique optimum is indeed interior. We show this by construction. For the p_0 and σ chosen above, both q_0 and q_1 lie strictly between zero and one. Their binomial masses are positive, so $m_0, m_1 > 0$. Furthermore, by the Fundamental Theorem of Calculus,

$$m_1 - m_0 = U'(1; p) - U'(0; p) = \int_0^1 U''(a; p_0)da < 0.$$

Hence (m_1, m_0) is a non-empty interval. Now choose, say $\gamma = (m_0 + m_1)/2$. Then, substitute γ back into $U'(0; p)$ and $U'(1; p)$, we obtain

$$U'(0; p) = \frac{m_0 - m_1}{2} > 0,$$

and

$$U'(1; p) = -\frac{m_0 - m_1}{2} < 0.$$

Thus, by the Intermediate Value Theorem, together with strict concavity, we have shown that a unique interior maximiser exists. Moreover, since p_0 was available for every admissible (n, k), the construction proves non-emptiness in every case.

Step 3: Show openness. Finally, we want to show the openness of the set to establish robustness beyond a potentially measure-zero set. Note that the second-order derivative is jointly continuous in inclination a and parameters. Define

$$F(\theta) = \max_{a\in[0,1]} U''(a; \theta).$$

To see its continuity at any $\theta_0 \in \mathcal{D}$, observe that

$$|F(\theta) - F(\theta_0)| \leq \sup_{a \in [0,1]} |U''(a;\theta) - U''(a;\theta_0)|.$$

As $\theta \to \theta_0$, the right-hand side tends to zero by uniform continuity of U'' on the Cartesian product of $[0,1]$ and a compact parameter neighbourhood of θ_0. Hence, continuity implies that the set $\{F < 0\}$ is open since the preimage of an open set under a continuous function is open. Moreover, since the endpoints m_0, m_1 are also continuous, we know that $\{m_1 - \gamma < 0\}$ and $\{\gamma - m_0 < 0\}$ are open for the similar reason. Thus, the set $\mathcal{O}_{n,k}$ can be written as:

$$\mathcal{O}_{n,k} = \{F < 0\} \cap \{m_1 - \gamma < 0\} \cap \{\gamma - m_0 < 0\}$$

which is open since finite intersections of open sets are open. We conclude that for every constructed parameter vector, there is a neighbourhood in which strict concavity, the opposing signs of the endpoint marginal incentives, and the unique interior maximiser persist, as required. □

Proof of Proposition 4. Let $N = n - 1$, and write

$$A(q) = T_{N,k}(q), \qquad C(q) = b_{N,k-1}(q), \qquad q = p + \sigma(a - p).$$

The objective is

$$U(a;p) = A(q) + aC(q) - \gamma a.$$

Its first derivative with respect to a is

$$U_a = \sigma A'(q) + C(q) + a\sigma C'(q) - \gamma.$$

Since $U_{aa}(a^*;p) < 0$ on $\mathcal{O}_{n,k}$, the implicit-function theorem applied to $U_a(a^*;p) = 0$ gives

$$\frac{\partial a^*}{\partial \sigma} = -\frac{U_{a\sigma}(a^*;p)}{U_{aa}(a^*;p)}.$$

Direct differentiation yields

$$U_{aa} = 2\sigma C'(q) + \sigma^2 \left[A''(q) + aC''(q)\right]$$

and

$$U_{a\sigma} = A'(q) + (2a - p)C'(q) + \sigma(a - p)\left[A''(q) + aC''(q)\right].$$

Combining these expressions gives

$$U_{a\sigma} = A'(q) + p\, C'(q) + \frac{a - p}{\sigma} U_{aa}.$$

Moreover, the binomial-tail derivative identities imply

$$A'(q) = N b_{N-1,k-1}(q), \qquad C'(q) = N\left[b_{N-1,k-2}(q) - b_{N-1,k-1}(q)\right].$$

Therefore,

$$\begin{aligned} A'(q) + p\, C'(q) &= N\left[(1-p) b_{N-1,k-1}(q) + p\, b_{N-1,k-2}(q)\right] \\ &= \Lambda(q,p) > 0. \end{aligned}$$

Evaluating at $a = a^*$ and substituting into the implicit-function formula gives

$$\frac{\partial a^*}{\partial \sigma} = \frac{\Lambda(q^*,p)}{-U_{aa}(a^*;p)} - \frac{a^* - p}{\sigma}.$$

If $a^* \leq p$, then the first term is strictly positive and the second is non-negative. Hence

$$\frac{\partial a^*}{\partial \sigma} > 0.$$

It remains to show that either sign is possible when $a^* > p$. Let

$$n = 3, \qquad k = 2, \qquad p = \frac{4}{5}, \qquad \sigma = \frac{1}{4}.$$

For these parameters,

$$U_{aa}(a; p) = -\frac{3 + 30a}{40} < 0 \qquad \text{for every } a \in [0, 1],$$

so any interior stationary point is the unique global maximiser. First, let

$$\gamma = \frac{237}{640}.$$

Direct substitution into the first-order condition gives

$$a^* = \frac{19}{20} > p.$$

At this optimum,

$$U_{a\sigma}(a^*; p) = \frac{49}{400}, \qquad U_{aa}(a^*; p) = -\frac{63}{80},$$

and therefore

$$\frac{\partial a^*}{\partial \sigma} = -\frac{U_{a\sigma}(a^*; p)}{U_{aa}(a^*; p)} = \frac{7}{45} > 0.$$

Next, let

$$\gamma = \frac{3299}{9600}.$$

The unique optimum is now

$$a^* = \frac{59}{60} > p.$$

At this optimum,

$$U_{a\sigma}(a^*; p) = -\frac{13}{1200}, \qquad U_{aa}(a^*; p) = -\frac{13}{16},$$

so

$$\frac{\partial a^*}{\partial \sigma} = -\frac{U_{a\sigma}(a^*; p)}{U_{aa}(a^*; p)} = -\frac{1}{75} < 0.$$

Both parameter vectors belong to $\mathcal{O}_{3,2}$. Since the inequalities are strict and a^* is continuously differentiable on $\mathcal{O}_{3,2}$, the two signs persist on non-empty open subsets. □

Proof of Proposition 5. We use subscripts to denote derivatives with respect to the corresponding argument. Fix an admissible threshold (n, k) and a parameter vector

$$(\sigma, p, \gamma) \in \mathcal{O}_{n,k}.$$

By Theorem 2, the objective is strictly concave on $[0, 1]$ and has a unique interior maximiser

$$a^*(\sigma, p, \gamma).$$

Hence

$$U_a(a^*; \sigma, p, \gamma) = 0, \qquad U_{aa}(a^*; \sigma, p, \gamma) < 0.$$

Since U is continuously differentiable in the parameters and twice continuously differentiable in a, the Implicit

Function Theorem implies that

$$a^* : \mathcal{O}_{n,k} \to (0,1)$$

is continuously differentiable. We first consider the normalised contribution cost γ. Differentiating the first-order condition with respect to γ gives

$$U_{aa}(a^*)\frac{\partial a^*}{\partial \gamma} + U_{a\gamma}(a^*) = 0.$$

Since the normalised cost enters the objective as $-\gamma a$,

$$U_{a\gamma} = -1.$$

Therefore,

$$\frac{\partial a^*}{\partial \gamma} = \frac{1}{U_{aa}(a^*)} < 0,$$

where the inequality follows from strict concavity. We next consider the social background p. Write

$$A(q) = T_{N,k}(q), \qquad C(q) = b_{N,k-1}(q),$$

so that

$$U(a;\sigma,p,\gamma) = A(q) + aC(q) - \gamma a, \qquad q = \sigma a + (1-\sigma)p.$$

Differentiating the first-order condition with respect to p yields

$$\frac{\partial a^*}{\partial p} = -\frac{U_{ap}(a^*)}{U_{aa}(a^*)}.$$

Since

$$U_{aa}(a^*) < 0,$$

it is sufficient to identify a region on which

$$U_{ap}(a^*) < 0.$$

Direct differentiation gives

$$U_{ap}(a;\sigma,p,\gamma) = (1-\sigma)\left[C'(q) + \sigma\left(A''(q) + aC''(q)\right)\right].$$

We use the same local construction as in the proof of Theorem 2. Choose

$$p_0 \in \left(\frac{k-1}{N}, 1\right).$$

Since the binomial mass $b_{N,k-1}$ is strictly decreasing to the right of its maximiser,

$$C'(p_0) < 0.$$

By continuity, there is a neighbourhood I of p_0 and some $\eta > 0$ such that

$$C'(q) \le -\eta \qquad \text{for all } q \in I.$$

Moreover, because A'' and C'' are continuous on $[0,1]$, there exists $M < \infty$ such that

$$|A''(q) + aC''(q)| \le M \qquad \text{for all } (a,q) \in [0,1]^2.$$

For $\sigma > 0$ sufficiently small,

$$q(a,p_0) = p_0 + \sigma(a - p_0)$$

lies in I uniformly for all $a \in [0,1]$. Taking σ smaller if necessary so that

$$\sigma M < \eta,$$

we obtain

$$C'(q) + \sigma\left(A''(q) + aC''(q)\right) < 0 \qquad \text{for all } a \in [0,1].$$

Hence

$$U_{ap}(a;\sigma,p_0,\gamma) < 0 \qquad \text{for all } a \in [0,1].$$

The non-emptiness construction in Theorem 2 also shows that, for such sufficiently small positive σ, one can choose γ so that

$$(\sigma,p_0,\gamma) \in \mathcal{O}_{n,k}.$$

At this parameter vector,

$$U_{ap}(a^*) < 0,$$

and therefore

$$\frac{\partial a^*}{\partial p} < 0.$$

Finally, U_{ap} and $a^*(\sigma,p,\gamma)$ are continuous on $\mathcal{O}_{n,k}$. The strict inequality

$$U_{ap}(a^*) < 0$$

therefore persists in a neighbourhood of the parameter vector constructed above. Consequently, the set

$$\mathcal{R}_{n,k} = \left\{(\sigma,p,\gamma) \in \mathcal{O}_{n,k} : U_{ap}\big(a^*(\sigma,p,\gamma);\sigma,p,\gamma\big) < 0\right\}$$

is open and non-empty. On $\mathcal{R}_{n,k}$,

$$\frac{\partial a^*}{\partial p} < 0.$$

□

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